Data-Driven Computationally Intensive Theory Development *

Nicholas Berente, *University of Georgia*
Stefan Seidel, *University of Liechtenstein*
Hani Safadi, *University of Georgia*

* This is a draft. The final version of this article has been published in *Information Systems Research*: Berente, N., Seidel, S., & Safadi, H. (2019). Research commentary—data-driven computationally intensive theory development. *Information Systems Research*, 30(1), 50-64.

Abstract

Increasingly abundant trace data provides an opportunity for information systems researchers to generate new theory. In this research commentary, we draw on the largely "manual" tradition of the grounded theory methodology and the highly "automated" process of computational theory discovery in the sciences to develop a general approach to computationally intensive theory development from trace data. This approach involves the iterative application of four general processes: sampling, synchronic analysis, lexical framing, and diachronic analysis. We provide examples from recent research in information systems.

Keywords: grounded theory methodology, computational theory discovery, GTM, computational, trace data, theory development, lexicon, inductive

1. Introduction

The abundant and ever-increasing digital trace data now widely available offer boundless opportunities for a computationally intensive social science (DiMaggio, 2015; Lazer et al., 2009). By "trace data" we refer to the digital records of activity and events that involve information technologies (Howison, Wiggins, & Crowston, 2011). Given the ubiquitous digitization of so many phenomena, some expect the widespread availability of a variety of trace data to do nothing less than revolutionize the social sciences and challenge established paradigms (Lazer, et al., 2009). Through direct computational attention to trace data, researchers can generate richer and more accurate understandings of social life—insights closer to the source (Latour, 2010).

Trace data typically requires computational tools for novel visualizations and pattern identification (Lazer, et al., 2009), which provides ample fodder for predictive modeling (Shmueli & Koppius, 2011). But such models, patterns, and visualizations are not *theory* (Agarwal & Dhar, 2014).

To unleash the power of trace data, information systems researchers can benefit from a general approach—a common ground across perspectives—for inductively generating novel theory from this data in all its forms. In this research commentary, we describe such an approach, rooted in the Grounded Theory Method (GTM—Glaser & Strauss, 1967), yet also informed by Computational Theory Discovery (CTD) in science fields. We propose a general, computationally-intensive approach to the inductive generation of theory from trace data. By describing the approach as "computationally-intensive," we seek to emphasize that it is neither classic, manual GTM nor entirely automated CTD. Instead, there is combination of manual and automated activity. The process involves key activities: sampling, synchronic analysis, lexical framing, diachronic analysis, and builds upon the key idea of emergence through iteration across these activities. We highlight the important role of the theoretical and "pre-theoretic" vocabulary, or lexicon, within which researchers frame the trace data in order to construct theory. Although the importance of a sense-making lexicon may seem obvious, it is important to appreciate the theoretically-loaded character of scholarly lexicons when generating theory from trace data. The choice of lexicon matters; it both enables and constrains the theoretical contribution that one can construct from trace data.

We thus look to extend the principles and spirit of GTM for alternative empirically-grounded inductive approaches that do not necessarily follow the prescriptions of GTM. This can perhaps make way for a new generation of methodological prescriptions specifically suited to computationally-intensive analysis of trace data and their combination with more traditional

forms of GTM analysis. Particularly in the context of widely available trace data and computational social science, the unprecedented access to different forms of data can drive novel inductive approaches that are consistent with the general approach of GTM, but perhaps not with existing, established methodological guidance.

We proceed as follows. In the next section, we define trace data and provide an overview. Then we highlight the role of lexicons in enabling and constraining theory development, and we compare "manual" grounded theory development and the "automated" process of computational theory discovery. Grounded in this analysis, we develop a general approach to computationally-intensive theory development. Our resulting framework is intended to guide empirically-grounded theory construction based on any kind of data using a variety of automated and manual techniques. We illustrate our approach with three published cases and we conclude by reviewing the contributions of this commentary.

2. Trace Data, Grounded Theory, and Computational Theory Discovery

When an activity or event occurs in conjunction with information technologies it often leaves a digital record, or "trace." The term "trace data" refers to digital records of activities and events that involve information technologies. Trace data is a form of unobtrusive measure (Webb et al 1966) that is enabled by digital technologies. Trace data is different from many other common forms of social science data in a variety of ways (Howison, et al., 2011). Often, trace data is "found" data—a byproduct of activities, not data generated for the purpose of research. In the case of qualitative data, for example, analysis of existing texts would involve trace data, whereas analysis of interview transcripts conducted for the research project would not. Further, trace data are "event-based" records of activities and transactions. Therefore, trace data is longitudinal and can take the form of time-stamped sequences of activities. Clickstreams, sensor data, and social media updates are all time-stamped sequenced trace data, but a cross-sectional record of, say, user attitudes or intentions is not.

By our definition, information systems researchers have been analyzing various forms of trace data for decades. Texts such as emails or other documents, transaction data from organizational systems, and social media updates are all forms of trace data. One might even conclude that the information systems field is particularly well-suited to the study of trace data (Agarwal & Dhar, 2014; Howison, et al., 2011). What is new, however, is the ever-increasing aspects of virtually every phenomenon that now leaves digital traces, and this is only expected to increase (Lazer, et al., 2009). A few decades ago trace data involved only data that was stored in a handful of organizational systems. Now the number and breadth of organizational systems has increased dramatically. In the past, a good deal of organizational activity occurred outside the purview of organizational systems. Given the widespread adoption of enterprise information systems, document and content management systems, advanced productivity applications, and others systems, most organizational activities now leave some sort of trace in terms of log files and communication or document trails. Further, devices are more abundant in organizational activity, including mobile phones, specialized mobile devices, various sensor and tracking technologies, and elements of the emerging Internet of Things. Outside of the organization, more and more people are using social media, mobile applications, and an ever-increasing number of sensors associated with the "digitized self"—homes, cities, and societies are all becoming sensitized.

Given this abundance of data, researchers can investigate a multitude of questions, increasing the number and variety of researchers who will investigate information systems phenomena through trace data. Certainly, traditional hypothetico-deductive (hypothesis testing) methods will continue to be a dominant approach to analyzing trace data, but the temptation to engage in open-ended exploration of this abundant data will also be strong to gain insight into a variety of phenomena. As information systems researchers look to generate theory from trace data, a common ground can be helpful to communicate across traditions.

3. Manual versus Automated Data-Driven Theory Development and the Role of the Lexicon

To develop a general approach to theory development from trace data, we highlight the importance of the researchers' lexicon in enabling and constraining a theoretical contribution, and we compare two polar traditions in inductive theory development—the intensely manual, largely qualitative, tradition of grounded theory methodology (GTM) and the highly automated, quantitative process of computational theory discovery (CTD). In doing so, we look for commonalities (see Figure 1).

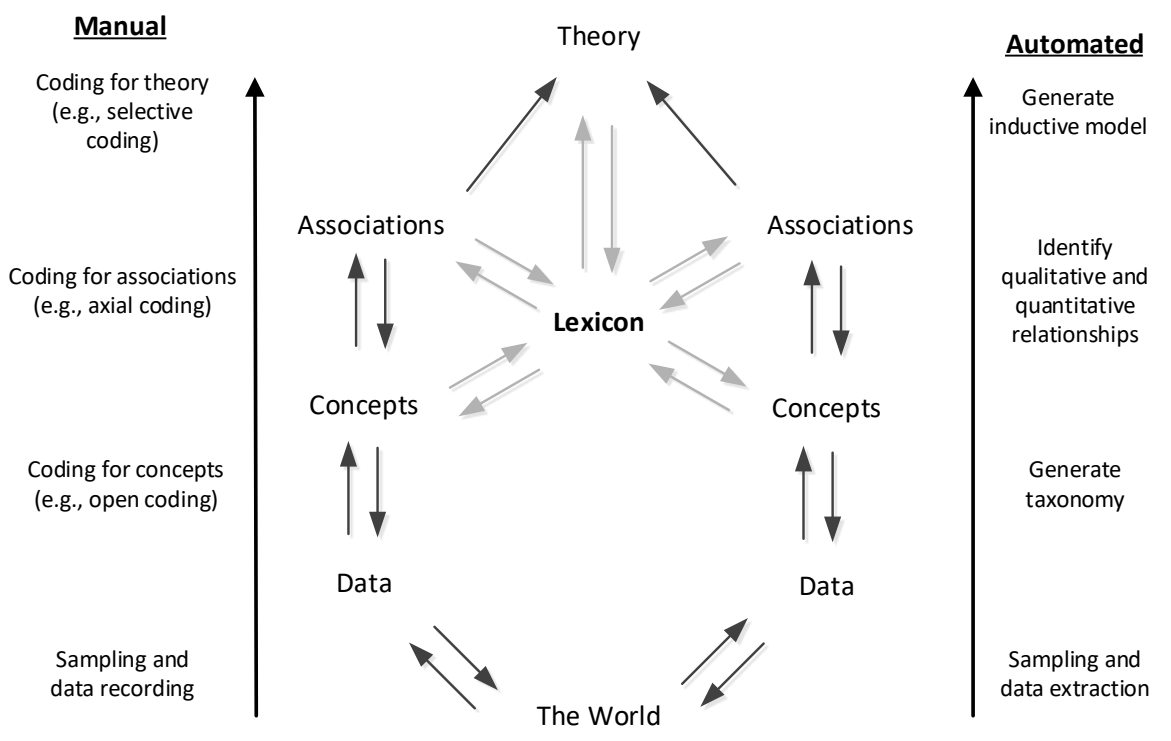

Figure 1. The role of the lexicon in "manual" and "automated" empirically-driven theory generation (black arrows describe the iterative process of building theory, with the general direction going from empirical to theoretical; gray arrows describe referencing to and from the lexicon in this process)

3.1 The Role of Lexicon in Developing Theory Grounded in Data

The language choices that researchers make are fundamental to any scientific endeavor and critically important for eliminating ambiguity in research and enabling research traditions to move forward (Podsakoff, MacKenzie, & Podsakoff, 2016). In his seminal work on organizational theory, Bacharach (1989) points out that "theory" is essentially a linguistic device that researchers use to organize empirical data in a way that simplifies those complex data with the use of concepts, and that asserts certain relationships among those concepts within some boundary condition and constraints.[1] Researchers construct theories through iterative, creative reasoning, but this theorizing does not occur from a "blank slate"—researchers necessarily draw upon prior scholarship in the theory construction process (Van de Ven, 2007).

In his philosophy of scientific knowledge, Juergen Habermas (1983, 2003) pointed out that different communities of researchers use language in very specific, theoretically-loaded ways. When analyzing empirical data through a theoretical lens, scientists use a lexicon shared by their community, which provides ready-made constructs and statements of relationships that they can then build upon. Habermas referred to a lexicon as the "pre-theoretic" grammar that is required for building any theoretical contribution. Situating scientific work in a lexicon both enables and constrains the scientific contribution. The lexicon enables because researchers do not have to reinvent all theoretical relationships from the ground up and the lexicon acts as a pre-theoretic basis for their contribution. The lexicon constrains the contribution because, in choosing a particular lexicon, scientists adopt the path dependent foundation that limits the degrees of freedom for their theoretical contribution.

The language researchers use or extend in their theorizing does not arise wholecloth, but language is always generated in relation to pre-existing theoretically-loaded language.

[1] Note that the definition of "theory" is a contested issue (see DiMaggio, 1995; Sutton & Staw, 1995; Weick, 1995), but to conceive of theory in terms of general statements about the relationship among concepts is a commonly accepted view (Jaccard & Jacoby 2010).

Thus, any new theorizing necessarily draws upon the lexicon of a particular scientific community. The lexicon is not a trivial issue of word choice, but in the scientific endeavor it is critical to enabling and constraining any contribution to knowledge. Next we describe the process of manual grounded theory development and then compare it with computational theory discovery, highlighting the role of the lexicon in each.

3.2 The Process of "Manual" Grounded Theory Methodology

Grounded Theory Methodology ("GTM," Glaser & Strauss, 1967) has been one of the strongest catalysts for widespread acceptance of qualitative research as well as inductive theory building across a variety of social science disciplines (Bryant & Charmaz, 2007; Eisenhardt, 1989). Grounded theory seeks to develop theoretical concepts and relationships while being informed by intense analysis of empirical data (Glaser & Strauss, 1967; Strauss & Corbin, 1990).

Over the years, GTM has evolved into a contested "family" of methodologies, rather than one very specific method (Bryant & Charmaz, 2007). This family of methods is replete with variants and rich in reflective discourse (Walsh, et al., 2015). There are disagreements on coding procedures (e.g., Kelle, 2007), the role of existing research (e.g., Jones & Noble, 2007), epistemological foundations (e.g., Charmaz, 2000), and a host of other divisions (also compare Seidel & Urquhart, 2013). From a unifying perspective, however, the method can be thought to involve building or extending a lexicon in a substantive area of investigation while, at the same time, drawing on an existing lexicon as pre-theoretic understanding in support of further sense-making with observations in the data.

Traditional "manual" GTM begins with the world's biggest dataset—the world itself—and reduces this dataset by sampling from the world in an area of interest. This sampling should be theoretical—what is known as "theoretical sampling"—in that the sample

should be developed and extended based on the results of analyzing that existing sample (Glaser & Strauss, 1967). In this view, a smaller, initial sample should be taken and analyzed, then subsequent samples should be informed by this analysis—they should help to follow-up on the insights that began to emerge from the initial sample. As such, the sample emerges over time, and this emergence is informed by existing analysis.

Coding and categorizing data is a fundamental activity in GTM, and many of the prescriptions for qualitative coding involve an intensely manual process (Charmaz, 2006; Goulding, 2002; Holton, 2007). These coding strategies may transfer directly to trace data, such as "trace ethnographies" (Geiger & Ribes, 2011) or "discourse archives" (Levina & Vaast, 2015) and some coding processes can likely be automated with machine learning, natural language processing, and other computationally-intensive techniques. Coding is not necessarily for only qualitative data but can also apply to quantitative data (Glaser, 2008).

The process of coding involves multiple passes through the data, iteratively identifying concepts and categories (i.e., more abstract concepts) that become more general at each pass, and then iteratively relating these concepts and categories to each other, resulting in the generation of theory. In the spirit of theoretical sampling, this analysis informs additional data collection, which then informs subsequent rounds of analysis—much the way a detective follows up on new leads given new information (Morse 2007). This continued sampling and analysis may involve various qualitative and quantitative data sources, including interviews (i.e., the coding of someone else's statements), observations such as online community threads, or memos written by the researcher throughout the analysis (Levina & Vaast, 2015). In GTM, coding is not something that happens after the data is collected, but occurs in interaction with the data collection, each informs the other, and coding is shaped by and in turn shapes different approaches to data collection. Codes reflect the constant comparison of emergent analysis

with existing bodies of knowledge and their respective lexicons. Thus, there cannot be any grounded theory development without a pre-theoretic lexicon, and the myth of the researcher as a 'blank slate' has been repeatedly debunked (Urquhart & Fernández, 2013). While the pre-theoretic lexicon is not necessarily applied in the sense of pre-conceived, a-priori concepts and relationships, it is drawn upon over the course of the research by the analyst to enhance her theoretical sensitivity in interactions with the field (Charmaz, 2006).

This process (see the left side of Figure 1) of manual GTM can be summarized in the following steps (see Appendix B for details of each step):

(1) *Initial sampling* from the world, then continued rounds of *theoretical sampling*, to record data
(2) Iterative coding to *identify concepts*, drawing on one or more lexicons
(3) Further coding and pattern matching to *identify associations* and relationships, again drawing on the salient lexicons
(4) Iterative sense-making of associations in relation to the pre-theoretic and theoretic understanding of existing lexicons in the relevant fields to *construct theory*

The data sample, the concepts and associations, the lexicon, and the resulting theory emerge from an intensely iterative process over time. Through coding and analyzing the data, the analyst moves from the descriptive to the conceptual level, and the result of this process are statements of relationships between concepts (Holton, 2007) that together constitute theory. This analysis process involves both synchronic (i.e., identification of concepts and associations in any given moment in time) and diachronic (i.e., identification of time dependent relationships between concepts, for instance, in terms of cause-effect relationships) approaches to analyzing data (Holland, Holyoak, Nisbett, & Thagard, 1986). Coding and analysis can follow a number of paths (Charmaz, 2006; Glaser, 1978, 1992; Strauss, 1987; Strauss & Corbin, 1990, 1998; Urquhart, 2013), the most well-known of which are the open, axial, and selective coding cycles in Straussian GTM (e.g., Strauss & Corbin, 1990, 1998). While there has been

intensive debate about—and disagreement on—the different coding strategies proposed in different approaches to GTM[2] (e.g., Bryant & Charmaz, 2007; Duchscher & Morgan, 2004; Matavire & Brown, 2011), all versions of grounded theory involve the four stages of sampling, identification of concepts, identification of associations, and the construction of an integrated theoretical scheme. This coding process is fundamental to GTM and is generally a manually-intensive process.

3.3 The Process of "Automated" Computational Theory Discovery

On a general level, the process of the grounded theory methodology has striking parallels to Computational Theory Discovery (CTD)—a discipline that emerged in the 1970s to automate the process of scientific research in the hard sciences to produce "discoveries" through artificial intelligence and machine learning techniques (Džeroski, Langley, & Todorovski, 2007). These techniques accompany a worldview that sees hypothetico-deductive methods as an artifact of computational limits that might be an outdated remnant of history:

> *"While the history of science can serve as an argument for norms of practice, for several reasons it is not a very good argument. The historical success of researchers working without computers, search algorithms, and modern measurement techniques has no rational bearing at all on whether such methods are optimal, or even feasible, for researchers working today. It certainly says nothing about the rationality of alternative methods of inquiry. Neither was nor is implies ought. The 'Popperian' method of trial and error dominated science from the sixteenth through the twentieth century not because the method was ideal, but because of human limitations, including limitations in our ability to compute"* (Glymour, 2004, pp. 74-75).

The history of science is rife with examples of discovering theories from observations, a process that modern epistemologists and scientists sought to understand. Herbert Simon proposed a view of theory discovery as heuristic problem solving. In this paradigm, scientists use mental operators to advance through a large search space from one knowledge state into another. Newell drew on this idea to provide a framework for both a theory of human problem solving and an approach to building computer programs with similar capabilities (Džeroski, et al., 2007). Computational theory discovery is rooted in this view and seeks to explicate human

[2] Glaser (1978, 1992), for instance, distinguishes the stages of open, selective, and theoretical coding.

intellect and construct a computational implementation of discovery processes (Wagman, 1997). Computational theory discovery (CTD) approaches have enjoyed successful outcomes in a variety of scientific fields including in mathematics, physics, chemistry, and genetics (Wagman, 2000).

In addition to CTD, various computational techniques for learning generalizable models from observations were developed in the discipline of machine learning. In particular, *Knowledge Discovery in Databases* (KDD) emerged in the 1990s to make sense of large transactional datasets by mapping low-level, hard-to-understand data to higher-level concepts (Fayyad, Piatetsky-Shapiro, Smyth, & Uthurusamy, 1996). Recently KDD received acceptability in the scientific community (Gaber, 2009). KDD and CTD share the same premise of using data to extract patterns and identify hypotheses (Williamson, 2009). Indeed, pioneers of the two disciplines pointed to their commonalities (Klösgen & Żytkow, 1996).

These computational disciplines share an underlying inductive framework and they are all geared towards extracting patterns from data and learning higher-level models and representations (Glymour, Madigan, Pregibon, & Smyth, 1996). Across a variety of fields "econometricians, statisticians, and data mining specialists are generally looking for insights that can be extracted from the data" (Varian, 2014, p. 5). In KDD, the progression from data to knowledge proceeds throughout the five steps of selection, preprocessing, transformation, data mining, and interpretation and evaluation (Fayyad, et al., 1996, p. 41). In CTD, Langley's (2000) model outlines four major steps, rooted in the three types of scientific knowledge that constitute the major products of the scientific enterprise (Džeroski, et al., 2007). Following Langley's model, the process of theory discovery (see the right side of Figure 1) can be summarized in the following steps aiming at generating scientific knowledge from observations:

(1) *Sampling observations* from phenomena of interest
(2) Iteratively *generate a taxonomy* of concepts from observations drawing on one or more lexicons

(3) *Identify qualitative and quantitative relationships* and associations among concepts of the taxonomy
(4) Iteratively *generate structural and process models* by drawing on associations in relation to the pre-theoretic and theoretic understanding of existing lexicons in the relevant fields

It is important to note that computational theory discovery is automated, but not automatic. Concepts are organized around people's theories about the world—the background knowledge and lexicons that guide and constrain learning (Wisniewski & Medin, 1995). There is a significant element of human interaction in all stages of the process. In particular, the role of humans is critical in the sampling process—in choosing which data to analyze and why. Humans choose a data sample to address a particular problem, and this problem formulation is a key element of any such analysis, and is inevitably an intensely human endeavor (Simon 1996). Humans interact with automating systems throughout the process—which often includes additional data collection and validation processes. Without intense human interaction, CTD projects can readily fail (Gaber, 2009). Computational theory discovery is not intended to supplant the role of the researcher, but to amplify it (Glymour, 2004, p. 77). Human knowledge has been shown to be superior to machine learning in identifying problematic cases for which data is scarce, and therefore the role of a human in even the most automated process cannot be underestimated (Attenberg, Ipeirotis, & Provost, 2015).

4. A Computationally-Intensive Approach to Theory Development

In this section, grounded in our analysis of both manual and automated theory discovery, we develop an approach that allows for different computationally-intensive grounded theory techniques ranging from predominantly manual theory development to predominantly automated theory discovery. Such abstraction is consistent with the idea of grounded theory as a meta theory allowing for all sorts of instantiations, where analysts combine different manual and automated methods (Walsh, 2015), with varying degrees of computational intensity.

A key insight from our analysis of these two approaches to inductive theory generation is that *all* research nowadays involves both manual and computational components. In the time that Glaser and Strauss conducted their path breaking research, the process of manual grounded theory generation took place primarily (if not entirely) on paper—but this is no longer the case. Researchers transcribe, code, and analyze their data using all sorts of qualitative and quantitative software tools. Nevertheless, we refer to this approach as "manual" to compare it with computational theory discovery techniques. Similarly, computational techniques are clearly not entirely automated but inevitably involve manual guidance and human judgement (Todorovski & Džeroski, 2007). Between these two approaches, there is a space for a host of combined, "computationally-intensive" techniques that offer fertile opportunity for theory generation (Figure 2).

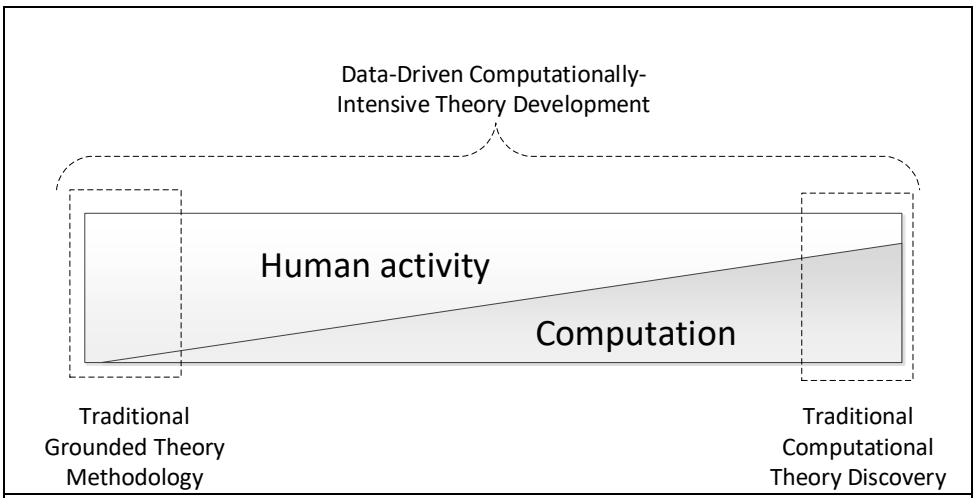

Figure 2. Data Driven Computationally-Intensive Theory Development - combining human & computational methods in varying proportions.[3]

Building upon the two poles of existing traditions for inductive theory generation—GTM and CTD—we can now propose an abstracted process for combined, computationally-intensive grounded analysis, focusing on the role of the lexicon in enabling the generation of theory from patterns identified in the data. Table 1 summarizes the main activities for both

[3] Note that the image is not symmetrical to indicate that, although there can be a purely manual grounded theory approach, there is no entirely computational approach to theory discover in that it inevitably involves human activity (the right side of the figure). The relative magnitude between these poles is for illustrative purposes only and is not intended to illustrate the relative space of one approach versus another in any way.

manual grounded and automatic computational theory generating processes—integrating the two approaches to highlight the potential for their interplay in a study.

It is important to emphasize that manual analysis and computational analysis are complements rather than substitutes. The researcher adopting a computationally-intensive grounded theory approach can integrate manual and computational analyses. For example, when sampling and collecting data, one might start with theoretical sampling via interviews and later identify opportunities to enrich the dataset with trace data—or vice-versa. In synchronic analysis, the researcher can classify trace data using codes identified manually or identify such codes computationally using clustering and validate them manually. Associations uncovered computationally are manually assessed for content validity. Manual associations between codes can benefit from a rigorous computational treatment. In diachronic analysis, the researcher may validate and quantify theories that were manually grounded or make sense of structural and process models that were computationally discovered. Theorizing is a process of sense-making and abstraction that demands human ingenuity and creativity. Computational methods can increase the efficiency and reliability of researchers by allowing them to examine vast quantities of data and consider various questions that can arise from the data simultaneously (Glymour, 2004, p. 77). Further, it is important to note that these activities do not follow a sequence in discrete steps, but iterate and emerge across the steps as the exploratory research unfolds. Following we provide a summary of each iterative step.

Table 1: Combining activities and goals in manual and automated analysis

Activity	**Goal**	**"Manual" Grounded Theory Methodology**	**"Automated" Computational Theory Discovery**
Sampling and data collection	Iteratively develop dataset	"Theoretical sampling"	"Recording observations"
		Combination: Iteratively constructing a dataset through cycles of data collection as a result of interaction with the phenomena of interest and the digital traces of those phenomena.	

Synchronic analysis	Categorize the data using concepts and identify associations among concepts	"Coding for concepts and associations" (e.g., open coding and axial coding in Straussian GTM)	"Create taxonomy" (e.g., using cluster analysis or association rule mining)
		Combination: Iteratively categorizing data according to established concepts and looking for qualitative and quantitative relationships and associations of these concepts to each other in the data.	
Lexical framing	Draw upon and extend the language of one or more research communities	"Codes and Relationships"	"Taxonomy and Associations"
		Combination: The lexicon provides the pre-theoretic reference for the naming of concepts and the identification of patterns in relation to a goal, using the language and causal relations determined by one or more scholarly communities.	
Diachronic analysis	Generate theory	"Constructing theory" (e.g., selective coding in Straussian GTM) "Theorizing" (e.g. temporal bracketing, grounded process theorizing)	"Develop model" (e.g., correlations, regressions and decision trees) Develop inductive model (e.g. process induction, process mining)
		Combination: The generation of theory requires a sense-making process. Rooted in empirical evidence, the analyst decides what concepts and relationships (pre-theoretic understanding) to include in elaborating a coherent theoretical scheme (theoretic understanding) and thus proposing an extension to the knowledge of a particular community of researchers.	

Sampling and data collection

At the outset, the analyst defines the area of investigation, thereby defining the scope and boundary conditions of the intended theory development. Often this begins by convenience—a dataset or study location is available; or by a phenomenon—some topic domain is "hot" at some point so researchers look to explore that domain. In early stages of research, an initial sample is drawn and analyzed. While in manual grounded theory the researcher often actively contributes to the process of data collection (e.g., through interviewing), trace data is typically "found" data (e.g., generated through user activity). The process of further sampling guided by insights from this first round of analysis ensues (Glaser & Strauss, 1967). In tradi-

tional grounded theory methodology, the sampling process ("theoretical sampling") is expected to be very focused, in part because of the cognitive limits of individuals. For example, pointing out the need for efficient sampling, Morse stated:

> *"Computer programs, while invaluable, merely assist in placing the data in the best possible position to aid the researcher's cognitive work; such programs cannot actually do the analysis for the researcher. It is for this reason that collecting too much data results in a state of conceptual blindness on the part of the investigator. Excessive data is an impediment to [GTM] analysis, and the investigator will be swamped, scanning, rather than cognitively processing, the vast number of transcripts, unable to see the forest for the trees, or even the trees for the forest, for that matter." (2007, p.233)*

This sentiment is probably quite accurate for a strictly "manual" approach for grounded theory. Individuals pouring over qualitative data need to do so in part by minimizing the dataset in the interests of efficiency. However, the moment one recognizes the analytic benefits of computational technologies, one can appreciate how an interplay of analyses between qualitative and trace data provides better opportunity for theorizing. The spirit of theoretical sampling remains—one begins with a convenience sample, and this sample can be intentional (like conducting interviews) or can involve inductive analysis of trace data. Based on initial findings, the researcher then samples additional data—either of the same type or of a complementary type. According to Gaskin and associates (2014) this mixed analysis of qualitative and computational data (for example) enables researchers to "zoom in and out" of phenomena—zooming in to get a rich understanding of elements of the data in context, and zooming out to look for and verify broader patterns. Combining sorts of data in the iterative sampling process helps researchers avoid merely "rationalizing" (Garud, 2015) what they see in terms of a particular perspective.

Synchronic analysis

In conjunction with the rounds of sampling, researchers continually explore the data. In manual grounded theory, this involves coding for both concepts and associations between

concepts, and in computational analysis this involves developing a taxonomy. Holland and associates (1986) describe the categorization and raw association of concepts in terms of "synchronic" regularities. In the earlier stages of manual grounded theory, the researcher aims to identify first categories based on the similarities between empirical indicators as well as first co-occurrences of categories. In open coding (Strauss & Corbin, 1998), for instance, the analyst identifies categories by grouping similar incidents found in the data under the same label. In axial coding (Strauss & Corbin, 1998) the analyst looks for other categories (sub-categories) that co-occur with this category.[4] That is, the analyst looks for both similarities (grouping) and correlations (co-occurrence of categories and their subcategories). In the computational analysis of trace data, both processes (identifying categories and identifying associations) involve a process of converging to synchronic relations using a variety of clustering techniques (Duda, Hart, & Stork, 2001; Friedman, Hastie, & Tibshirani, 2001). Clustering associates observations in data to clusters based on their similarity. Observations in the same clusters share recurrent patterns or synchronic associations. Very often, the challenge for constructing synchronic associations is the exponential number of such relationships (Glymour, et al., 1996, p. 39). CTD focuses on finding a parsimonious, understandable, and communicable sets of relationships (Schwabacher & Langley, 2001). Identifying synchronic relations need not be either qualitative or computational, but can be both (Anderberg, 1973; Hipp, Güntzer, & Nakhaeizadeh, 2000).

Lexical framing

Iterating with the coding and continued sampling (as necessary), the analyst settles upon the lexical frameworks to be used to analyze the data—that is, the pre-theoretic lexicon

[4] Note that in axial coding the analyst also starts to identify whether the categories are indeed conditions or consequences, and the lines between synchronic and diachronic analysis blur. We illustrate our model by using GTM terminology borrowed from Strauss and Corbin (1990, 1998). In Glasarian GTM, synchronic analysis would primarily comprise of open and selective coding, where the former identifies first categories and the latter groups categories further (Glaser, 1978).

providing an appropriate grammar for analyzing the data. We refer to this conscious activity of drawing on a lexicon to attribute meaning to codes as "lexical framing" (Fillmore, 1976). This can, of course, involve drawing on multiple lexicons, but typically addresses that of the focal community of researchers, or "conversants" (Huff 1999). In the analysis process, the researcher may consider different pre-theoretic lexicons throughout the process, and the lexicon-in-use may change. Further, certain lexicons may be more or less abstract, which can influence the scope of emergent theorizing. Different levels of abstraction can be combined, as well. For example, one can draw upon abstract pre-theoretic lexicons such as the *coding paradigm* including labels such as "conditions," "actions/interactions", and "consequences" (Strauss & Corbin, 1998), and combine this with very specific classification schemes for computational analysis, like labeled or curated data sets for training learning algorithms.

Similarly, one can explore a dataset using multiple forms of cluster analysis techniques (Anderberg, 1973), but use *conceptual clustering* to supervise the algorithm by drawing on specific theoretical discourse, allowing the researcher to incorporate the desired aspects of categories that is independent of the data (Fisher, 1987; Michalski, 1980). Conceptual clustering involves using known attributes to categorize data and mimics human concept learning where concept formation relies on prior knowledge (Thompson & Langley, 1991).

Diachronic analysis

Since theory involves identifying causal or sequential patterns, theory development implicitly or explicitly involves a temporal element—and therefore theory construction necessarily rests on diachronic, temporal analyses (Holland, et al., 1986). Even in trace data, which necessarily has a temporal character, diachronic regularities are not necessarily self-evident. Although the digital traces are temporally ordered, issues like simultaneity, recurrence, and recursion can render the temporal interpretation of patterns ambiguous. Therefore, it is not

only the terminology in a lexicon that offers guidance, but so too do the time-ordered relationships established among the concepts as a starting point for temporal analysis.

In manual grounded theory, axial coding aims to relate concepts to each other, to support the end-goal that often involves identifying explanations in the form of cause and effect relationships. Selective coding then aims to integrate those explanations in relation to one core category representing the phenomenon studied, thereby producing an integrated theoretical scheme. In computational analysis, establishing diachronic relationships is achieved with some sort of inductive model. One form of inductive model is referred to as a "structural model,"[5] which relates concepts into quantitative laws in the form of rules, equations, and models (Rose & Langley, 1986). Various computational techniques exist to favor parsimonious explanations and uncover causal relationships from data (Pearl, 2011), and resulting structural models are refined and validated by researchers (Saito & Langley, 2007). Process models, another form of inductive model, focus on the time-dependent relationships among concepts rather than their stable over time associations. Concepts are often treated as states rather than variables, and ordering rather than correlation is used to relate them (Mohr, 1982). Several techniques are available for grounded process theorizing from longitudinal data (Langley, 1999; Van De Ven & Poole, 1995). For example, temporal bracketing—one common technique—is used to distinguish different phases over which the phenomenon of interest unfolded and to analyze how actions of one phase lead to changes in the context that will affect action in subsequent phases (Langley, 1999). Both computational and manual grounded theory analysis require a process of sense-making; data analysis is ultimately a cognitive human process (Grolemund & Wickham, 2014).

[5] Note that the term "structural model" has a somewhat different meaning in different fields. Here we use it to describe abstract models of stable relationships among variables.

While, longitudinal trace data spanning years and decades is easily obtained, it is missing the wider context that shapes relationships. Reconstructing context and relating it to emergent theory from data is one challenge given the overwhelming volume of data (Levina & Vaast, 2015). Computational techniques such as process induction and process mining offer to extract temporal relationships such as ordering and sequencing from trace data (Bridewell et al., 2008; Günther & Van Der Aalst, 2007). For example, Lindberg, Berente, Gaskin, & Lyytinen (2016) use process modeling to gain an inductive understanding of how developers in open-source communities resolve software code interdependencies over time. Similarly, recent advances in social network analysis allow for understanding generative mechanisms that lead to a sequence of events based on past patterns of events (Butts, 2008; Quintane, Conaldi, Tonellato, & Lomi, 2014). By focusing on the temporal dimension, these techniques extend knowledge established by structural models.

All aspects of computationally-intensive theory development emerge throughout the research project. The researcher acts as a sort of detective that finds a lead in the data and then pursues that lead looking at a variety of data sources using a variety of methodologies to construct valid theoretical propositions, while drawing on the lexicon of an existing community of researchers to move the understanding of those researchers forward. This approach is not entirely new, but it has been our goal to make it explicit in a general way for a general information systems audience.

5. Illustrations

We draw on three recent studies in information systems to illustrate the applicability of this approach to computationally-intensive theory development (Miranda, Kim, & Summers, 2015; Lindberg, et al., 2016; Vaast, Safadi, Lapointe, & Negoita, 2017). See Table 2 for a summary, followed by brief descriptions of each study.

Table 2: Illustrations of computationally-intensive theory development [Manual aspects in brackets]

	Miranda, Kim, & Summers (2015)	**Lindberg, Berente, Gaskin, & Lyytinen (2016)**	**Vaast, Safadi, Lapointe, & Negoita (2017)**
Sampling[6]	[decision to explore corporate use of social media and IT innovation] - round 1: 2,414 texts [decision to filter data to focus on early adopter firms] - round 2: 1,183 initiatives panel	[decision to explore Rubinius project] - round 1: 686 pull requests with 3,707 activities [manual examination of text attached to sequences] - round 2: 432 text excerpts	[decision to explore microblogging around oil spill] - round 1: 23,000 tweets [decision to focus data collection on three connective action episodes] - round 2: 1882 tweets
Synchronic Analysis	[manual coding: civic, domestic, industrial, inspiration, market, and renown] Automated coding [sense-making of cluster analysis] Categories of Vision: Efficiency- Engineer, Brand-Promoter, Good-Citizen and Master-of- Ceremonies [sense-making of network and temporal representation of data] Facets: coherence, continuity, clarity and diversity	Initial (activity) codes: assigned, closed, commented, mentioned, merged, opened, referenced, reopened, reviewed [referring to routine literature lexicon] Constructs: developer and development interdependencies; order and activity variation; [manual coding] Qualitative codes: Diagnosing, Causal theorizing, Asking for clarification, Clarification, Teaching, Adding features, Increasing code clarity, Increasing code functionality, Asking for tests, Providing tests, Asking for documentation, Providing documentation. [axial coding] Qualitative categories: knowledge integration; direct implementation	Episodes: Boycott BP, Stop the Drill, Hair and Fur [sensemaking of cluster analysis] Group clusters: advocates, supporters and amplifiers [sensemaking of time series and motif analysis] Enacted role characteristics: roles, frequency, intensity, pattern of feature use, actions, reciprocal interdependence
Lexical Framing	Organizing vision and diffusion of innovation theory; Orders of worth	Coordination and organizational routines theory; coordination in online communities	Organizational interdependency theory; affordance theory
Diachronic Analysis	Facets of different visions associated with differential diffusion	Process theory of coordination around unresolved interdependencies through direct implementation or knowledge integration	Theory of the role of artifacts in different connective actions
Benefit of combining manual and computational analyses	Automation	Complementarity	Creation of alternative representations

Miranda et al. (2015) develop theory on how different facets of organizing visions influence the diffusion of IT innovations in companies. To do so, they applied supervised content analysis, network visualization, and statistical analysis combined with traditional content analysis. Their research question was shaped by authors' interest in institutional theory and

[6] Note that in each of the illustrations sampling was an iterative process that was more complex than shown here. We briefly discuss this point when summarizing the three studies at the end of this section.

unpacking institutional mechanisms at play in organizations. The study involved two inductive stages: (1) extraction of mental schema and hierarchical structure of organizing visions from archival documents; and (2) exploration of facets of organizing visions in the diffusion of IT innovations. Throughout the process the researchers iterated across multiple rounds of sampling and analysis. The initial sample involved 46 of the Fortune 50 firms. The sample included text from social media, product descriptions, and other media outlets. Collectively this resulted in 2,414 text documents. In a second stage, the authors deliberately refined the sample to focus on a longitudinal panel of 1,183 initiatives that the researchers uncovered through manual analysis of the texts. Initial manual coding was subsequently automated through content analysis of the texts for presence of six principles: civic, domestic, industrial, inspiration, market, and renown, drawn from the "orders of worth" lexicon. The six principles served as dimensions of texts from which the authors sought to extract schemas of organizing vision using relational class analysis (RCA) that revealed four clusters, which were validated and labelled as: Efficiency-Engineer, Brand-Promoter, Good-Citizen and Master-of-Ceremonies. After identifying the four schemas the authors continued to investigate how the different visions affect diffusion with the sample of the 1,183 initiatives. They characterize differences in the schemas with four facets: coherence, continuity, clarity, and diversity by considering the schema variation over time. They then correlated the number of initiatives representing diffusion with the four facets. Visually examining the correlation scatter plots, the authors found out that some of these relationships are linear while others are quadratic. From this analysis, they theorize that organizing visions are hierarchies of schemas and that different facets in this hierarchy differentially drive the diffusion of IT innovation.

Lindberg, et al. (2016) explore an open source software community (Rubinius) to understand how community developers coordinate complex work in ways that go beyond arms-

length coordination mechanisms. They mix sequence and statistical analysis with manual coding and visual interpretation of data to develop a process theory for coordinating around unresolved interdependencies in such communities. Initially they sampled 686 pull requests across 12 months of an open source software project that included 3,704 activities. Initial rounds of computational analysis involved sequences and sequence covariates of timestamped activities labelled in the software development platform (GitHub activities: assigned, closed, commented, mentioned, merged, opened, referenced, reopened, reviewed). This analysis led to initial identification of pull requests that were variably complex. In referencing the lexicon from coordination and organizational routines theory applied to software development, they characterized this complexity in terms of "unresolved" developer and development interdependencies, and activity and order variation in the routines. They used combinations of regression and visual inspection to identify associations among types of interdependencies and routine variation. The final elements of their theory generation involved manual, qualitative analysis of a sample of 432 "text excerpts" from these complex pull requests. They qualitatively coded the second dataset a traditional GTM approach through multiple rounds of coding (final codes: diagnosing, causal theorizing, asking for clarification, clarification, teaching, adding features, increasing code clarity, increasing code functionality, asking for tests, providing tests, asking for documentation, providing documentation; final categories: knowledge integration; direct implementation). They concluded with a process theory of coordinating unresolved interdependencies in online communities through the mechanisms of knowledge integration and direct implementation.

Vaast et al. (2017) combine grounded theorizing with clustering, network motif analysis and time series analysis to examine how social media use affords new forms of organizing and collective engagement. The paper explores an oil spill in the Gulf of Mexico to understand new forms of collective engagement that they refer to as "connective action." Given this

focus, the authors decided to sample data from the microblog service Twitter. The study began with an initial sample of 23,000 tweets related to the Deepwater Horizon incident in April 2010 to broadly gain insight into microblogging activity in the wake of a disaster. The choice of this crisis was deliberate - because of its magnitude the crisis led to various forms of collective action. On Twitter, this was "the most microblogged issue in 2010."(p.1184) From a first round of manual open coding, three threads of communication they described as Connective Action Episodes (CAEs: Boycott BP, Stop the Drill, Hair and Fur) emerged. This observation then informed subsequent sampling. A second round of sampling focused on extracting all tweets related to the CAEs through trackbacks of originally identified tweets. Based on this refined theoretical sample, they conducted a new round of "manual" open coding focusing on the similarities and differences between the CAEs and resulted in a number of role categories. They switched to "automated" taxonomy creation to identify role categories by performing a cluster analysis using DBSCAN algorithm on their patterns of Twitter usage rather. They then looked at how members of these clusters participated in the three CAEs to contrast among episodes. The paper focuses on two types of associations: CAEs with actor categories longitudinally with temporal analysis, and actor categories within CAEs cross-sectionally using social network motif analysis. The temporal relationships are visualized with time-series plots to identify patterns. These patterns reflect interdependencies among actor categories. By characterizing the type of interdependence among actors in CAEs, they drew on different organizational theories of coordination and interdependency. Integrating the new lexicon with the theory of affordances, the authors introduce a theory of the role of connective affordances in the context of connective action.

The combination of manual and computational approaches allowed the authors of the three papers to go beyond what could have been achieved using only traditional methods. While the data collected in Miranda et al. (2015) lends itself to manual coding, identifying and

tracing organization visions over a long period of time is extremely challenging and perhaps not feasible. The supervised content analysis approach allowed the researchers to create categories based on their pre-theoretic understanding of the phenomenon and their exploratory analysis of a subsample. The benefit of the computational content analysis was to automate the lengthy and tedious process of manually coding the six-year data of fifty companies. In Lindberg et al. (2016), the collected ssequences from open-source repository have textual elements including description and comments by software developers. While sequence data lend itself naturally to computational analysis, textual elements are better understood with human sense-making. The two methods are complementary. Finally, Vaast et al. (2017)'s exploratory manual analysis of collected data led the researchers to focus on connective affordances of social media. Understanding the interconnections of a large number of people is a cognitively challenging task. The value of the computational methods was to complement manual sense-making to create alternative representations to understand connective action at scale. The network motif analysis provided a visual summary for researchers to interpret.

We verified these insights by communicating with all authors of these three papers to understand their own views of the challenges in applying computationally-intensive approaches. Authors highlighted a number of challenges of which the foremost involved identifying the appropriate reference lexicon. Authors used expressions such as "tying together the different visualizations with a coherent theoretical narrative" (Lindberg) and "connecting the data to a theoretical anchor in a way that made sense conceptually and that respected the collected data" (Vaast) as the key challenges. Further, they pointed out that there is no straightforward, mechanistic way to enact this approach. It is an intensely iterative and creative process that had no specific guidelines. Although the iteration between the sample and the analysis is often downplayed in the reporting, authors of all three studies indicated an intensely emergent process of sampling decisions and continuous analysis. As one author mentioned,

"the method we adopted was fairly emergent and idiosyncratic, it was not easy for us to refer to established guidelines for mixed-methods research… they could only provide templates that did not fully fit what the study was doing" (Miranda). All authors indicated that the bulk of the visualizations used along the way to develop their stories never made it into the final version of the paper. The authors pointed to open and innovative reviewers and editors who helped them to construct their stories in a convincing way. Finally, authors noted that things seem to be changing. They find that there is an increasing variety of tools (such as new R packages) and an increasing appreciation of computationally intensive approaches. As one author put it: "The tone in our community is increasingly one of accepting that computational tools will be important even to qualitative scholars and those focused on theory development" (Lindberg).

6. Discussion and Implications

> *Everyone who publishes in professional journals in the social sciences knows that you are supposed to start your article with a theory, then make deductions from it, then test it, and then revise the theory. At least that is the policy that journal editors and textbooks routinely support. In practice, however, I believe that this policy encourages—in fact demands, premature theorizing and often leads to making up hypotheses after the fact—which is contrary to the intent of hypothetico–deductive method.* (Locke, 2007, p. 867)

In Edwin Locke's quote, he points out how the policy of presenting research in terms of constructing hypotheses and testing them often goes against the real work of theory generation and empirical analysis. Quite frequently findings are developed inductively, and then researchers disingenuously reconstruct a hypothetico-deductive paper out of the patterns they found (Anonymous, 2015). This is because there is a stigma associated with "fishing" in data for patterns that may simply be spurious correlations without theoretical explanations. Post hoc rationalizations that justify results after the fact are to be avoided—and for good reason (Garud, 2015; Walsh, 2014). In the age of computational social science and trace data, there should be a mechanism for this pretense to come to an end. How can researchers inductively

generate theory from patterns they see in data, without feeling the need to repackage their research in terms of hypothesis testing? There is an important place for inductive theory generation, but researchers must be honest about what they are doing (Garud, 2015). The answer lies in general approach that highlights the role of lexicons in the emergent process of analyzing multiple forms of data—including trace data—for the purpose of generating theory.

For decades, qualitative information systems researchers have understood that rigorous attention to empirical data via cycles of sense-making can help generate novel theory. Using GTM, qualitative researchers have a legitimizing tradition to draw upon when explaining patterns they see in accordance with existing lexicons and proposing the resulting ideas in terms of theory generation (Walsh, 2014). We have highlighted the relevance of lexical framing in the process of identifying both concepts and relationships between concepts, and thus to facilitate theory emergence. Glaser and Strauss led a revolution of sorts in social analysis. Through a program of intense attention to empirical data, they legitimized a way to generate novel theory that could revitalize a stale discourse. Some argue that organizational and information systems literature may be stagnating (Davison, 2010) or not reaching their potential (Grover, 2013). Now, particularly given the opportunity that the data explosion provides, it is time to open up approaches for theory generation that are grounded in empirical data. At the same time, it is important to capitalize on the maturity and flexibility of GTM, and to encourage further methodological attention in this regard in order to get the most out of on the new opportunities proffered by the availability of trace data. Against this backdrop, we join with those calling for a broader "grounded paradigm" for theory development based on the key features of grounded theory (Walsh, et al., 2015). This paradigm is characterized by two key elements: the "grounded" and the "theory." *Grounded* refers to the intense attention to empirical data, comprised of rounds of sampling and analysis using a variety of qualitative, quantitative, and computational techniques. *Theory* refers to the patterns of associations that emerge from

this analysis as researchers construct their understandings of the phenomena by drawing on and extending the lexicon of a community of researchers. In this paper, we have sketched a broad approach for IS researchers dealing with any type of data using computationally-intensive approaches to theory development.

It has been recently stated that information mining and traditional theory building are indeed complementary, interrelated methods (Dhar, 2013; Gopal, Marsden, & Vanthienen, 2011). Data and knowledge mining methods, by themselves, however do not move forward the understanding of a phenomenon. To move this understanding forward, we need to theorize and explain the patterns of association that we identify. In order to make sense of patterns identified through computational methods, and to form appropriate mental models that can be used in the sense-making process (Holland, et al., 1986), the analyst requires a lexicon that is shared by a community of scholars (Habermas, 2003). This lexicon can be taken from existent theoretic lexicons, such as the social network perspective, that, in turn, serve as pre-theoretic lexicons in the process of novel theorizing. Similarly, the patterns generated through computational analysis constitute pre-theoretic understanding in form of synchronic regularities that can serve as a foundation for the development of novel theory. Our framework can thus be seen as an answer to the call made by Gopal et al. (2011), who suggest that "researchers may develop an iterative approach that uses information mining outcomes as inputs into the theory construction and validation processes" (p. 370). Overall, theory developed from a combination of techniques can be more robust than theory generated from a single qualitative dataset, as researchers triangulate and cycle through different approaches (Van de Ven, 2007). The general approach to computationally intensive theory development accommodates different combinations of manual and automated activities. It draws attention to the opportunity afforded by the widespread abundance of trace data, and finds that the interplay of manual and computational techniques together can drive novel theorizing and is entirely consistent with

GTM, but is also open to other forms of computationally-intensive inductive analysis. This approach is just a start and more work is needed to flesh it out. Other should push the grounded paradigm further.

The information systems field is particularly well-positioned to lead a methodological revolution in computationally-intensive social research (Agarwal & Dhar, 2014). As a discipline, we investigate those phenomena that have made the trace data revolution possible in the first place. Our discipline is devoted to investigating complex sociotechnical settings that require us to make sense of large amounts of data that pertain to the interaction of 'the social' and 'the technical' (Orlikowski, 2007). Further, there is a very real need to develop novel and accurate theory grounded in large amounts of data instead of continually "working" existing theories (Legewie & Schervier-Legewie, 2004), as we are challenged to further develop our intellectual core. The importance of new methodological approaches cannot be underestimated. Some of the most important, innovative, Nobel-prize winning findings owe themselves to methodological advancements (Greenwald, 2012). If one were to compare this trace data opportunity in social science to physics, "it is as if every physicist had a supercollider dropped into his or her backyard" (Davis, 2010, p. 696), and the field of information systems is poised to contribute.

7. Conclusion

> *On the one hand, there isn't a paradigm for data scientists to easily publish their inductive data discoveries. These, often highly insightful, findings either go unpublished or are turned into hypotheses followed by testing to suit mainstream publication requirements. On the other hand, grounded theory scholars are increasingly encountering large digitized data archives that cannot be reasonably analyzed with qualitative methods alone. Thus we would all benefit if we start including inductive data scientists into the grounded theory research community and start using some of the advanced analytical techniques available today.* (Levina in Walsh, et al., 2015, p. 11)

In her quote above, Levina points to the opportunity presented by the abundance of trace data, and argues for incorporating computational analyses in our empirically grounded theory development efforts. In this research commentary, we inquired into how the lessons learned from GTM can be used to build theory from trace data, thereby building a general

framework for this approach. Specifically, we highlight the importance of a lexicon in this process. It is perhaps noteworthy that the development of our approach itself has indeed grounded components, too. We looked to what, at first blush, may be described as polar extreme approaches to theory development—the automated CTD tradition and the GTM approach. In relating the two we find that there is quite a bit of similarity at a general level of abstraction, and we develop a general approach based on this similarity.

Armed with this general approach, we encourage researchers to act like detectives when looking to generate theory. It is important to note that this is not a methodology, per se, but a general approach whereby researchers creatively use qualitative approaches as needed, but do so in conjunction with a variety of computational techniques—ever employing new techniques as they come online—to triangulate and validate insights and conjectures, resulting in potentially more robust and creative theorizing. Their detective work, however, cannot ignore the cumulative knowledge of the community of scientists, and it is critical to highlight the role of lexicons as the source of and destination for that knowledge.

References

Agarwal, R., & Dhar, V. (2014). Editorial—Big Data, Data Science, and Analytics: The Opportunity and Challenge for IS Research. *INFORMATION SYSTEMS RESEARCH, 25*(3), 443-448.

Anderberg, M. R. (1973). Cluster analysis for applications: DTIC Document.

Anonymous. (2015). The Case of the Hypothesis That Never Was; Uncovering the Deceptive Use of Post Hoc Hypotheses. *Journal of Management Inquiry*, 1056492614567042.

Attenberg, J., Ipeirotis, P., & Provost, F. (2015). Beat the Machine: Challenging Humans to Find a Predictive Model's "Unknown Unknowns". *Journal of Data and Information Quality (JDIQ), 6*(1), 1.

Bacharach, S. B. (1989). Organizational theories: Some criteria for evaluation. *Academy of Management Review, 14*(4), 496-515.

Birks, D. F., Fernandez, W., Levina, N., & Nasirin, S. (2013). Grounded theory method in information systems research: its nature, diversity and opportunities. *European Journal of Information Systems, 22*(1), 1-8.

Bridewell, W., Langley, P., Todorovski, L., Dzeroski, S., Džeroski, S., & Todorovksi, L. (2008). Inductive process modeling. *Machine Learning, 71*, 1-32. doi: 10.1007/s10994-007-5042-6

Bryant, A., & Charmaz, K. (2007). Grounded Theory Research: Methods and Practices. In A. Bryant & K. Charmaz (Eds.), *The Sage handbook of grounded theory* (pp. 1-28). London, UK: Sage.

Butts, C. T. (2008). A relational event framework for social action. *Sociological Methodology, 38*, 155-200. doi: 10.1111/j.1467-9531.2008.00203.x

Charmaz, K. (2000). Grounded theory: Objectivist and constructivist methods. In N. K. Denzin & Y. S. Lincoln (Eds.), *Handbook of qualitative research* (2nd ed., pp. 509–535). Thousand Oaks, CA: Sage.

Charmaz, K. (2006). *Constructing grounded theory: A practical guide through qualitative analysis*. Thousand Oaks, CA: Sage.

Davis, G. F. (2010). Do theories of organizations progress? *Organizational Research Methods*.
Davison, R. M. (2010). Retrospect and prospect: information systems in the last and next 25 years: response and extension. *Journal of Information Technology, 25*(4), 352-354.
Dhar, V. (2013). Data science and prediction. *Communications of the ACM, 56*(12), 64-73.
DiMaggio, P. J. (1995). Comments on" What theory is not". *Administrative Science Quarterly*, 391-397.
DiMaggio, P. J. (2015). Adapting computational text analysis to social science (and vice versa). *Big Data & Society, 2*, 205395171560290. doi: 10.1177/2053951715602908
Duchscher, J. E., & Morgan, B. (2004). Grounded theory: Reflections on the emergence vs forcing debate. *Journal of Advanced Nursing, 48*(6), 605–612.
Pattern Classification, John Wiley & Sons (2001).
Džeroski, S., Langley, P., & Todorovski, L. (2007). Computational Discovery of Scientific Knowledge. In S. Džeroski & L. Todorovski (Eds.), *Computational Discovery of Scientific Knowledge: Introduction, Techniques, and Applications in Environmental and Life Sciences* (pp. 1-14). Berlin, Heidelberg: Springer Berlin Heidelberg.
Eisenhardt, K. M. (1989). Building Theories from Case Study Research. *Academy of Management Review, 14*, 532-550. doi: 10.5465/AMR.1989.4308385
Fayyad, U. M., Piatetsky-Shapiro, G., Smyth, P., & Uthurusamy, R. (1996). Advances in knowledge discovery and data mining.
Fillmore, C. J. (1976). Frame semantics and the nature of language. *Annals of the New York Academy of Sciences, 280*(1), 20-32.
Fisher, D. H. (1987). Knowledge acquisition via incremental conceptual clustering. *Machine Learning, 2*, 139-172. doi: 10.1007/BF00114265
Friedman, J., Hastie, T., & Tibshirani, R. (2001). The elements of statistical learning. *1*.
Gaber, M. M. (2009). Scientific data mining and knowledge discovery.
Garud, R. (2015). Eyes wide shut? A commentary on the hypothesis that never was. *Journal of Management Inquiry, 24*(4), 450-454.
Gaskin, J., Berente, N., Lyytinen, K., & Yoo, Y. (2014). Toward Generalizable Sociomaterial Inquiry: A Computational Approach for 'Zooming In & Out' of Sociomaterial Routines. *MIS Quarterly, 38*(3), 849-871.
Geiger, R. S., & Ribes, D. (2011). *Trace ethnography: Following coordination through documentary practices.* Paper presented at the System Sciences (HICSS), 2011 44th Hawaii International Conference on.
Glaser, B. G. (1978). *Theoretical Sensitivity: Advances in the Methodology of Grounded Theory*. Mill Valley, CA: The Sociology Press.
Glaser, B. G. (1992). *Basics of grounded theory analysis: Emergence vs. forcing*. Mill Valley, CA: Sociology Press.
Glaser, B. G. (2008). *Doing Quantitative Grounded Theory*: Sociology Press.
Glaser, B. G., & Strauss, A. L. (1967). *The discovery of grounded theory: Strategies for qualitative research*. Chicago, IL: Aldine Publishing Company.
Glymour, C. (2004). The automation of discovery. *Daedalus, 133*(1), 69-77.
Glymour, C., Madigan, D., Pregibon, D., & Smyth, P. (1996). Statistical inference and data mining. *Communications of the ACM, 39*(11), 35-41.
Gopal, R., Marsden, J. R., & Vanthienen, J. (2011). Information mining—Reflections on recent advancements and the road ahead in data, text, and media mining. *Decision Support Systems, 51*(4), 727-731.
Goulding, C. (2002). *Grounded theory: A practical guide for management, business and market researchers*: Sage.
Greenwald, A. G. (2012). There is nothing so theoretical as a good method. *Perspectives on psychological science, 7*(2), 99-108.
Grolemund, G., & Wickham, H. (2014). A Cognitive Interpretation of Data Analysis. *International Statistical Review*.
Grover, V. (2013). Muddling Along to Moving Beyond in IS Research: Getting from Good to Great. [Article]. *Journal of the Association for Information Systems, 14*, 274-282.
Günther, C. W., & Van Der Aalst, W. M. P. (2007). Fuzzy mining–adaptive process simplification based on multi-perspective metrics. *International Conference on Business Process Management*: Springer.
Habermas, J. (1983). Interpretive social science vs. hermeneuticism. *Social science as moral inquiry*, 251-269.
Habermas, J. (2003). *Truth and justification*. Cambridge, MA: MIT Press.
Hipp, J., Güntzer, U., & Nakhaeizadeh, G. (2000). Algorithms for association rule mining—a general survey and comparison. *ACM sigkdd explorations newsletter, 2*(1), 58-64.
Holland, J. H., Holyoak, K. J., Nisbett, R. E., & Thagard, P. R. (1986). Induction: Processes of inference, learning, and discovery: MIT Press, Cambridge, MA.

Holton, J. A. (2007). The coding process and its challenges. *The Sage handbook of grounded theory*, 265-289.
Howison, J., Wiggins, A., & Crowston, K. (2011). Validity issues in the use of social network analysis with digital trace data. *Journal of the Association for Information Systems, 12*(12), 767-797.
Jaccard, J., and Jacoby, J. *Theory Construction and Model-Building Skills* Guilford Press, New York, 2010.
Jones, R., & Noble, G. (2007). Grounded theory and management research: a lack of integrity? *Qualitative Research in Organizations and Management: An International Journal, 2*(2), 84-103.
Kelle, U. (2007). The development of categories: Different approaches in grounded theory. In A. Bryant & K. Charmaz (Eds.), *The Sage Handbook of Grounded Theory* (pp. 191-213). London, UK: Sage.
Klösgen, W., & Żytkow, J. M. (1996). *Knowledge discovery in databases terminology.* Paper presented at the Advances in knowledge discovery and data mining.
Kuhn, T. S. (1962). *The Structure of Scientific Revolutions,* : University of Chicago Press.
Langley, A. (1999). Strategies for theorizing from process data. *Academy of Management Review, 24*, 691-710. doi: 10.5465/AMR.1999.2553248
Langley, P. (2000). The computational support of scientific discovery. *International Journal of Human-Computer Studies, 53*(3), 393-410.
Latour, B. (2005). Reassembling the social-an introduction to actor-network-theory. *Reassembling the Social-An Introduction to Actor-Network-Theory, by Bruno Latour, pp. 316. Foreword by Bruno Latour. Oxford University Press, Sep 2005. ISBN-10: 0199256047. ISBN-13: 9780199256044, 1*.
Latour, B. (2010). Tarde's idea of quantification. In M. Candea (Ed.), *The Social After Gabriel Tarde: Debates and Assessments*: Routledge.
Lazer, D., Pentland, A. S., Adamic, L., Aral, S., Barabasi, A. L., Brewer, D., . . . Gutmann, M. (2009). Life in the network: the coming age of computational social science. *Science, 323*(5915), 721.
Legewie, H., & Schervier-Legewie, B. (2004). *Research is hard work, it's always a bit suffering. Therefore on the other side it should be fun.* Paper presented at the Anselm Strauss in conversation with Heiner Legewie and Barbara Schervier-Legewie. Forum Qualitative Sozialforschung/Forum: Qualitative Social Research.
Leveraging archival data from online communities for grounded process theorizing (Routledge 2015).
Lindberg, A., Berente, N., Gaskin, J., & Lyytinen, K. (2016). Coordinating Interdependencies in Online Communities: A Study of an Open Source Software Project. *Information Systems Research*, isre.2016.0673. doi: 10.1287/isre.2016.0673
Locke, E. A. (2007). The Case for Inductive Theory Building†. *Journal of Management, 33*(6), 867-890.
Matavire, R., & Brown, I. (2011). Profiling grounded theory approaches in information systems research. *European Journal of Information Systems, 22*(1), 119-129.
Michalski, R. S. (1980). Knowledge acquisition through conceptual clustering: A theoretical framework and an algorithm for partitioning data into conjunctive concepts. *Journal of Policy Analysis and Information Systems, 4*, 219-244.
Miranda, S. M., Kim, I., & Summers, J. D. (2015). Jamming with Social Media: How Cognitive Structuring of Organizing Vision Facets Affects IT Innovation Diffusion. *Mis Quarterly, 39*, 591-614.
Mohr, L. B. (1982). Explaining Organizational Behavior.
Morse, J. (2007). Sampling in grounded theory. The Sage handbook of grounded theory, 229-244.
Orlikowski, W. J. (2007). Sociomaterial practices: Exploring technology at work. *Organization Studies, 28*(9), 1435-1448.
Pearl, J. (2011). Statistics and causality: Separated to reunite-commentary on Bryan Dowd's "separated at Birth". *Health Services Research, 46*, 421-429. doi: 10.1111/j.1475-6773.2011.01243.x
Podsakoff, P. M., MacKenzie, S. B., & Podsakoff, N. P. (2016). Recommendations for creating better concept definitions in the organizational, behavioral, and social sciences. *Organizational Research Methods, 19*(2), 159-203.
Quintane, E., Conaldi, G., Tonellato, M., & Lomi, A. (2014). Modeling Relational Events: A Case Study on an Open Source Software Project. *Organizational Research Methods, 17*, 23-50. doi: 10.1177/1094428113517007
Rose, D., & Langley, P. (1986). Chemical discovery as belief revision. *Machine Learning, 1*(4), 423-452.
Quantitative Revision of Scientific Models, 4660 120-137 (Springer Berlin Heidelberg 2007).
Schwabacher, M., & Langley, P. (2001). *Discovering Communicable Scientific Knowledge from Spatio-Temporal Data.* Paper presented at the Proceedings of the Eighteenth International Conference on Machine Learning.
Seidel, S., & Urquhart, C. (2013). On emergence and forcing in information systems grounded theory studies: The case of Strauss and Corbin. *Journal of Information Technology, 28*(3), 237-260.
Shmueli, G., & Koppius, O. R. (2011). Predictive analytics in information systems research. *MIS Quarterly, 35*(3), 553-572.

Strauss, A. L. (1987). *Qualitative analysis for social scientists*. Cambridge, UK: University of Cambridge Press.
Strauss, A. L., & Corbin, J. (1990). *Basics of Qualitative Research* (1st edition ed.). Thousand Oaks, CA: Sage.
Strauss, A. L., & Corbin, J. (1998). *Basics of qualitative research. Techniques and procedures for developing grounded theory* (2nd ed.). London, UK: Sage.
Sutton, R. I., & Staw, B. M. (1995). What theory is not. *Administrative Science Quarterly*, 371-384.
Thompson, K., & Langley, P. (1991). Concept formation in structured domains. *Concept formation: Knowledge and experience in …*. doi: 10.1016/B978-1-4832-0773-5.50011-0
Integrating domain knowledge in equation discovery 69-97 (Springer 2007).
Urquhart, C. (2013). *Grounded theory for qualitative research: A practical guide*. London, UK: Sage.
Urquhart, C., & Fernández, W. (2013). Using grounded theory method in information systems: the researcher as blank slate and other myths. *Journal of Information Technology, 28*(3), 224-236.
Urquhart, C., Lehmann, H., & Myers, M. D. (2010). Putting the 'theory' back into grounded theory: Guidelines for grounded theory studies in information systems. *Information Systems Journal, 20*(4), 357-381.
Vaast, E., Safadi, H., Lapointe, L., & Negoita, B. (2017). Social Media Affordances for Connective Action - An Examination of Microblogging Use During the Gulf of Mexico Oil Spill. *MIS Quarterly, forthcoming*.
Van de Ven, A. H. (2007). *Engaged scholarship: a guide for organizational and social research: a guide for organizational and social research*. Oxford: Oxford University Press.
Van De Ven, A. H., & Poole, M. S. (1995). Explaining Development and Change in Organizations. *Academy of Management Review, 20*, 510-540. doi: 10.2307/258786
Varian, H. R. (2014). Big Data: New Tricks for Econometrics. *Journal of Economic Perspectives, 28*, 3-28. doi: 10.1257/jep.28.2.3
Wagman, M. (1997). General Unified Theory of Intelligence: Its Central Conceptions and Specific Application to Domains of Cognitive Science.
Wagman, M. (2000). Scientific discovery processes in humans and computers: Theory and research in psychology and artificial intelligence.
Walsh, I. (2014). Using grounded theory to avoid research misconduct in management science. *Grounded Theory Review, 13*(1).
Walsh, I. (2015). Using quantitative data in mixed-design grounded theory studies: An enhanced path to formal grounded theory in information systems. *European Journal of Information Systems, 24*, 531-557.
Walsh, I., Holton, J. A., Bailyn, L., Fernandez, W., Levina, N., & Glaser, B. G. (2015). What Grounded Theory Is . . . A Critically Reflective Conversation Among Scholars. *Organizational Research Methods*.
Webb, E., Campbell, D. T., Schwartz, R. D., and Sechrest, L. *Unobtrusive Measures: Non-Reactive Research in the Social Sciences* Sage Publications (Sage Classics Edition, Original Publication, 1966), Thousand Oaks, CA, 2000.
Weick, K. E. (1995). What Theory is Not, Theorizing Is. *Administrative Science Quarterly, 40*(3), 385-390.
Wisniewski, E. J., & Medin, D. L. (1995). Harpoons and long sticks: The interaction of theory and similarity in rule induction. *Goal-driven Learning*, 177.